Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

May 31, 2018

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. This problem asks you to analyze an asset-pricing model with time-varying net supply of an asset. There is measure one of identical agents who maximize their expected discounted utility of consumption. Each agent's instantaneous utility is given by $U(c)$, where c denotes consumption, and the function U is strictly increasing, concave, and continuously differentiable. The time discount factor is $\beta \in (0, 1)$.

There is one asset that yields a sequence of dividends $\{z_t\}_{t=0}^{\infty}$, where $z_t = z > 0$ is *constant* for all t. The total number of perfectly divisible equity shares of the asset is normalized to *two* (for simplicity). In each period t, external investors exogenously (or inelastically, i.e., regardless of the asset price) demand a certain quantity $\hat{y}_t$ of the shares. The remaining quantity, $\bar{y}_t \equiv 2 - \hat{y}_t$, is the *net supply* of the asset (which, in equilibrium, will be held by the agents in the economy). Assume that $\hat{y}_t = 0$ (so that $\bar{y}_t = 2$) if t is even and $\hat{y}_t = 1$ (so that $\bar{y}_t = 1$) if t is odd. Each consumer's initial endowment of the asset is $\bar{y}_{-1} = 1$.

As in the standard asset-pricing model, in the beginning of each period the agents receive dividends on the shares that they own from the previous period. Then they trade the current consumption good and asset shares at competitive markets. Prices are normalized such that the price of the current consumption good is one.

(a) Formulate an individual's recursive problem in this economy. Derive the first-order condition and the Envelope condition. Write down the Euler equation(s) evaluated at the competitive equilibrium quantities (and prices). *(Hint: Be careful identifying the aggregate state variable and substituting the market-clearing condition!)*

(b) Suppose that U is linear. How do the asset prices in odd and even periods compare? What are they equal to? Explain intuition behind your results.

(c) Suppose now that U is strictly concave. How do the asset prices in odd and even periods compare now? *(Hint: Compare the Euler equations in this case with those in part (b).)*

(d) Carefully explain intuition for the comparison of prices in part (c), and also why the comparison is different from that in part (b).

2. Consider a three-period OLG model. Time is discrete and continues forever, indexed by $t \geq 0$. The economy is populated by overlapping generations of individuals with finite lifetimes. Each individual lives for three periods: young in period 1, middle-aged in period 2, and old in period 3. In each period, the size of the population is constant and equals $N = N^y + N^m + N^o = 1$, where N^y, N^m, and N^o are the constant fraction of young, middle-aged, and old in the population. Assume that $N^o = (1-\delta)N^m = (1-\delta)^2 N^y$, where $\delta \in [0,1)$ is the constant death probability of young and middle-aged.

Preferences are described by

$$U(c_t^y, c_{t+1}^m, c_{t+2}^o) = \ln c_t^y + \beta \ln c_{t+1}^m + \beta^2 \ln c_{t+2}^o,$$

where $\beta \equiv 1-\delta$ is the survival probability of young and middle-aged, and c_t^y, c_{t+1}^m, and c_{t+2}^o are consumption of young, middle-aged, and old, respectively. Middle-aged individuals are endowed with one unit of time that they supply inelastically to a competitive labor market at the wage rate w_t. The lifetime budget constraint is

$$c_t^y + \frac{c_{t+1}^m}{R_{t+1}} + \frac{c_{t+2}^o}{R_{t+1}R_{t+2}} \leq \frac{w_{t+1}}{R_{t+1}},$$

where R_{t+1} and R_{t+2} are the gross interest rates between periods t and $t+1$, and $t+1$ and $t+2$, respectively. Output, Y_t, is produced with a Cobb-Douglas production function, $Y_t = A_t K_t^\alpha L_t^{1-\alpha}$, where $A_t = A_0(1+g)^t$ grows over time at rate $g \geq 0$ with $A_0 \equiv 1$. K_t and L_t are capital and labor, respectively. Initial capital stock is K_0. The capital stock fully depreciates after one period, such that $K_{t+1} = I_t$ where I_t denotes investment.

(a) Define the competitive equilibrium of this OLG economy.

(b) Derive the first-order conditions (FOCs) of the individual's problem. Using the FOCs, obtain the life-cycle profile of consumption (c_t^y, c_{t+1}^m, c_{t+2}^o) as a function of w_t, R_{t+1}, and R_{t+2}. Using (c_t^y, c_{t+1}^m, c_{t+2}^o), obtain the life-cycle profile of savings (s_t^y, s_{t+1}^m, s_{t+2}^o).

(c) Obtain the aggregate savings of the economy as $S_t = S_t^y + S_t^m + S_t^o$, where S_t^y, S_t^m, and S_t^o are total savings of the young, middle-aged, and old, respectively.

(d) Show that the law of motion of the aggregate capital-to-labor ratio, $k_t \equiv K_t/L_t$, takes the form $k_{t+1} = \text{constant} \times A_t k_t^\alpha$. Find such a constant. *(Hint: Use the market clearing condition $I_t = S_t$.)*

(e) Obtain the steady-state growth rate of the capital-to-labor ratio.

(f) Without performing derivations, explain intuitively whether the saving rate, S_t/Y_t, is increasing, decreasing, or constant in the death probability δ.

3. Consider an agent who has access to a natural resource—e.g., lumber—and can use it to produce a consumption good. The agent's lifetime utility is

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where c_t is consumption in period t, $\beta \in (0,1)$, and $u(\cdot)$ is a concave, increasing and continuously-differentiable function defined on $\mathbb{R}_+$ that satisfies the Inada conditions. Let x_t be the stock of lumber in the beginning of period t, $x_0 > 0$ is given. In the beginning of each period, the agent decides how much lumber l_t to use for production. The unused lumber grows at a constant rate g, implying

$$x_{t+1} = (x_t - l_t)(1+g).$$

The agent has access to a production technology that converts l_t units of lumber into $f(l_t)$ units of the consumption good, where $f(\cdot)$ is a concave, increasing, continuously-differentiable function defined on $\mathbb{R}_+$, and $f(0) = 0$. The consumption good is non-storable, implying that $c_t = f(l_t)$.

(a) Set up the agent's decision problem sequentially. Derive the Euler equation. Under which condition(s) on the primitives of the model does the solution to the agent's decision problem have a steady state?

(b) For the rest of the problem, assume that $u(c_t) = \ln(c_t)$ and $f(l_t) = l_t^{\gamma}$, where $\gamma \in (0,1)$. Find the agent's optimal consumption profile in closed form. Under which conditions is it constant over time? Is your answer consistent with the sufficient condition(s) for the existence of the steady state derived in part (a) above?

(c) Now suppose that there are two agents in the economy, who differ only in the growth rates of the lumber stocks they own. The growth rates are g_1 and g_2, where $g_2 > g_1$. Assume that the natural resource is not tradable, but in period 0 the two agents can trade promises to deliver the consumption good to each other in any period $t \geq 0$. Denote by d_t^i, $i = 1, 2$, the amount of claims to consumption in period t that agent i buys. Set up the sequential decision problem for agent i in period 0. Prove that trade must occur in equilibrium (e.g., there exist t and i such that $d_t^i \neq 0$). Without formal derivations, state which of the following two conditions must hold: $d_0^1 > 0$ or $d_0^2 > 0$. Explain your answer intuitively.